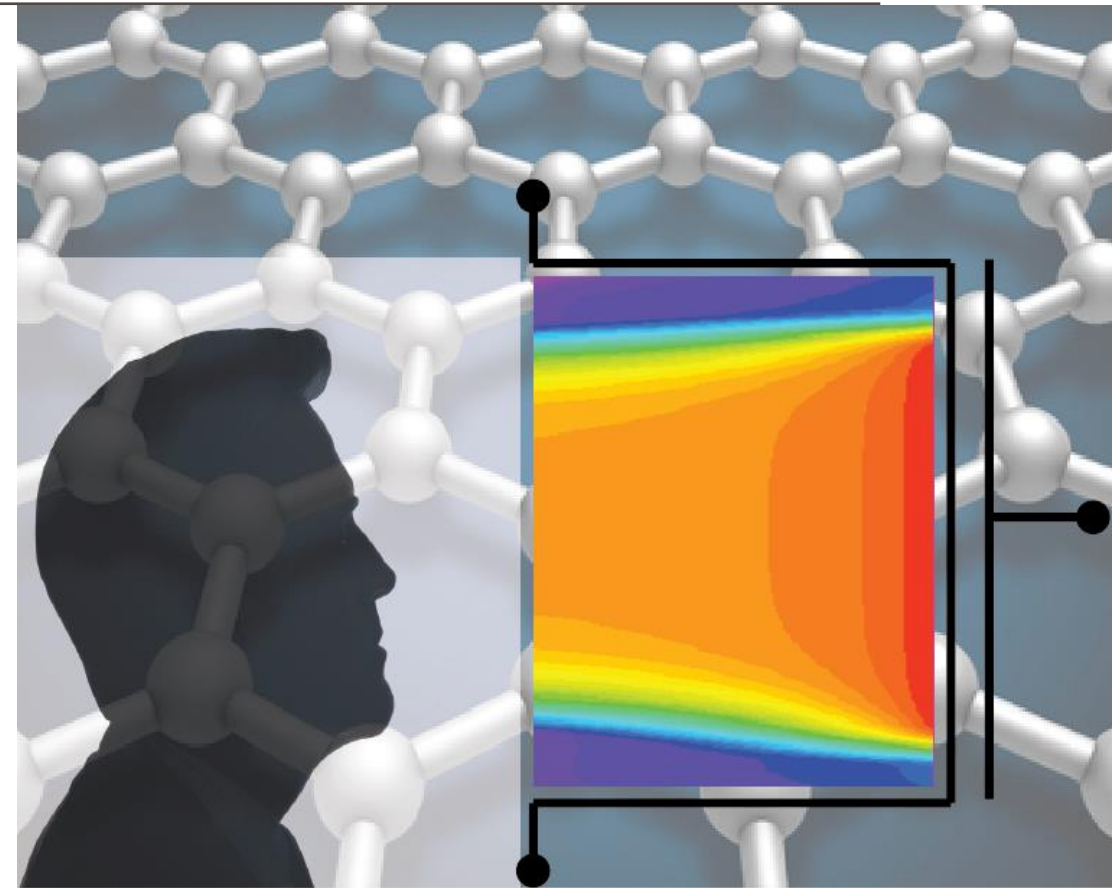


ADVANCED SOLID-STATE DEVICES

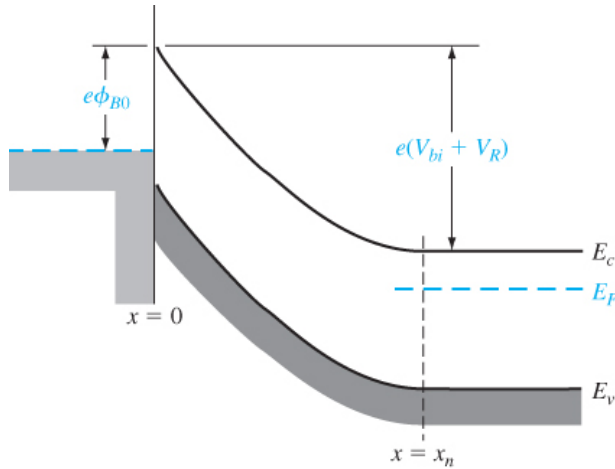
Week-6, Lecture-2

Sneh Saurabh
13th February, 2025



Metal—Semiconductor Junction

Rectifying MS Junction: Ideal Properties



- Using Poisson's Equation (and approach used for PN junction), we can derive the electric field, depletion region width, and junction capacitance
- Derive yourself

- Electric field at position x in the depletion region: $E = -\frac{qN_d}{\epsilon_s}(x_n - x)$
- Width of depletion region $W = x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{qN_d} \right]^{1/2}$
- Junction capacitance $C' = \left[\frac{\epsilon_s q N_d}{2(V_{bi}+V_R)} \right]^{1/2}$

Metal-Semiconductor Junction: Problem

Consider a contact between Tungsten and n-type semiconductor with $N_d = 10^{16} \text{ cm}^{-3}$ at equilibrium at 300 K. The work function of tungsten is $\phi_m = 4.55 \text{ V}$ and the electron affinity of silicon is $\chi = 4.01 \text{ V}$.

Determine the space charge width at zero bias.

[Take $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$ and $N_c = 2.8 \times 10^{19} \text{ cm}^{-3}$]

$$W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{qN_d} \right]^{1/2}$$

$$\begin{aligned}\phi_{B0} &= 0.54 \text{ V} \\ \phi_n &= 0.206 \text{ V} \\ V_{bi} &= 0.334 \text{ V}\end{aligned}$$

$$\begin{aligned}x_n &= \left[\frac{2 \times 11.7 \times 8.85 \times 10^{-14} \times 0.334}{1.6 \times 10^{-19} \times 10^{16}} \right]^{1/2} \\ &= 0.208 \times 10^{-4} \text{ cm}\end{aligned}$$

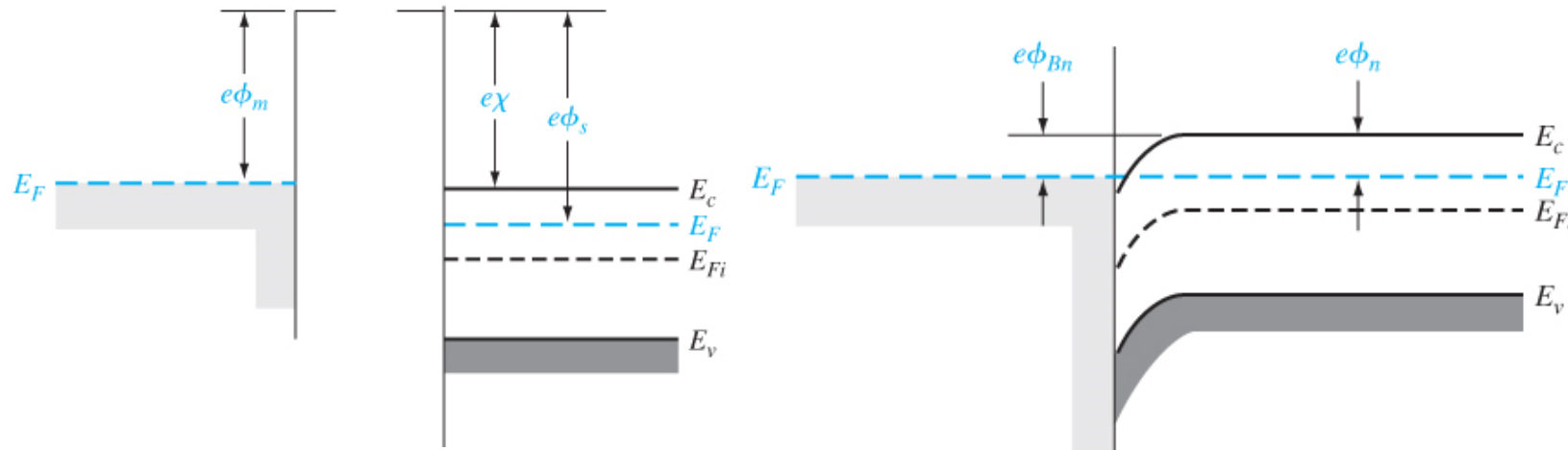
Metal—Semiconductor Junction



Non-rectifying

Non-rectifying MS Junction: Equilibrium

Problem: The band diagram of metal and n-type semiconductor when they are far apart is shown below. Assume $\phi_m < \phi_s$. Draw the band diagram when they are in contact.



- Effective barrier for electrons:
 $\phi_{Bn} = \phi_n$ (small for moderately or highly-doped semiconductor)

- At equilibrium, Fermi energy will align
- To align Fermi energy, electrons will flow from metal to the semiconductor because they are at higher energy
- Semiconductor surface will become richer in electron

- Conduction band edge will move closer to the Fermi level at surface (bend downward) on the semiconductor side
- No depletion region (surface charge in both metal and semiconductor)

Non-rectifying MS Junction: Under Bias

- On applying bias, almost all voltage appears across the semiconductor
- Semiconductor bands slope upward or downward depending on the bias

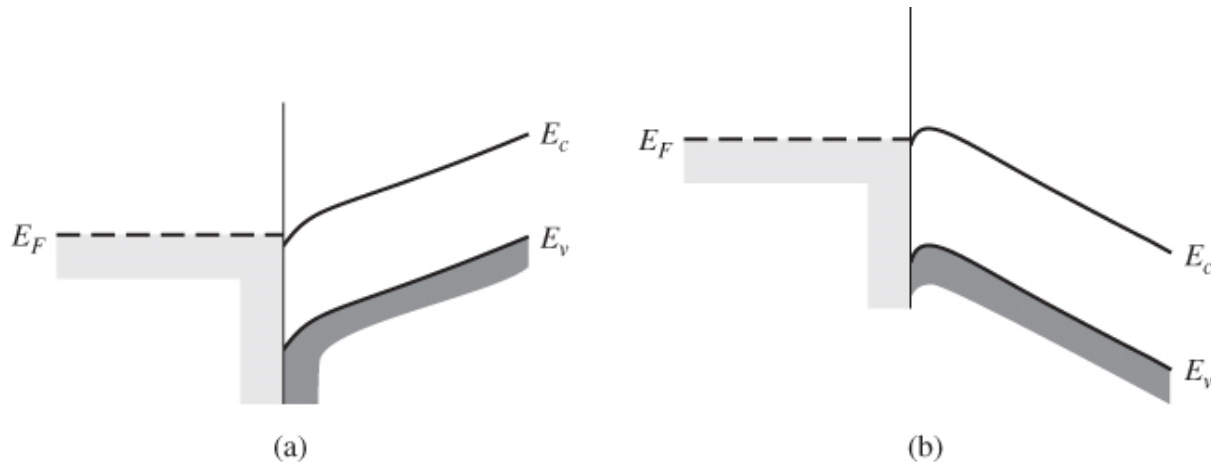


Figure 9.12 | Ideal energy-band diagram of a metal-n-type semiconductor ohmic contact (a) with a positive voltage applied to the metal and (b) with a positive voltage applied to the semiconductor.

Thus, barrier is non-rectifying in this case

Semiconductor at -ve potential with respect to metal (applying +ve voltage to metal):

- Band edge slopes upward
- Electrons free to move from semiconductor to metal (move down the slope)

Semiconductor at +ve potential with respect to metal (applying -ve voltage to metal):

- Band edge slopes downward
- Electrons free to move from metal to semiconductor (overcome small barrier and move down the slope)

Metal—Semiconductor Junction



Non-ideal Effects

Non-ideal Effects

- Schottky barrier height is a function of the *surface states* and *electric field*
 - Surface states are difficult to predict (need to know the microscopic structure of MS junction and interfacial reaction of metal and semiconductor atoms)
 - Schottky barrier height must be experimentally determined parameter

Metal—Semiconductor Junction



Current-Voltage Relationship

MS Junction: Mechanisms of Current Flow

Two types of current flow typically dominate in MS junction:

1. Thermionic Emission:

- Emission of electrons *over* the barrier from semiconductor to metal
- Common for moderately doped semiconductor. Example: Si with $N_D \leq 10^{17} cm^{-3}$ at room temperature

2. Quantum-mechanical Tunneling:

- Movement of electrons *through* the barrier
- Common for heavily doped semiconductor and responsible for Ohmic contact

- Both mechanisms involve current transport due to majority carriers
- Note that in PN junction diode the major contribution was due to minority carriers

MS Junction: Thermionic Emission (1)

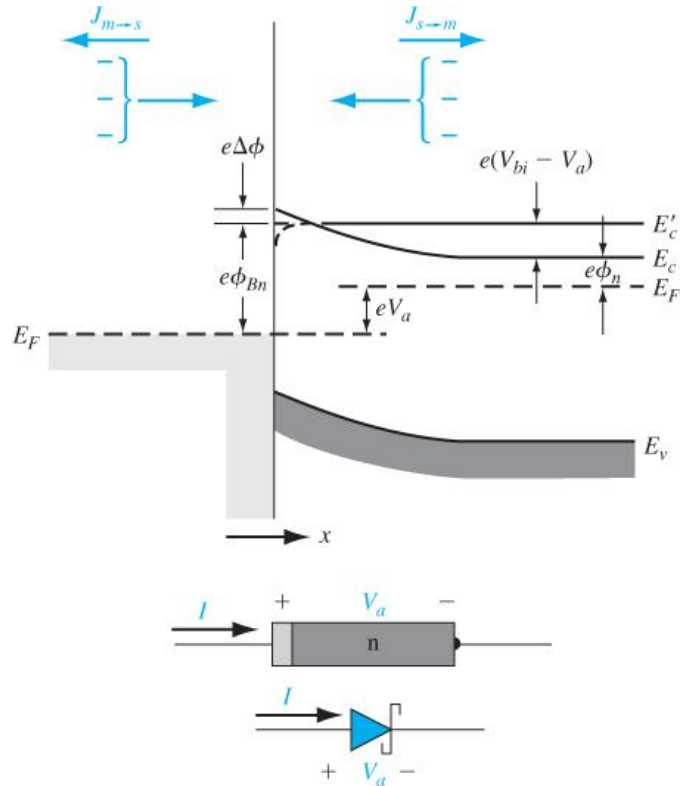


Figure 9.7 | Energy-band diagram of a forward-biased metal–semiconductor junction including the image lowering effect.

- Electrons that have energy greater than the barrier can cross it ($E > E'_c$)

$$J = J_{s \rightarrow m} - J_{m \rightarrow s}$$

$$= \left[A^* T^2 \exp \left(-\frac{\phi_{Bn}}{kT/q} \right) \right] \left[\exp \left(\frac{V_a}{kT/q} \right) - 1 \right]$$

- Parameter A^* is called effective Richardson constant

- Current flows: $J_{m \rightarrow s}$ and $J_{s \rightarrow m}$

MS Junction: Thermionic Emission (2)

$$J = \left[A^* T^2 \exp \left(-\frac{\phi_{Bn}}{kT/q} \right) \right] \left[\exp \left(\frac{V_a}{kT/q} \right) - 1 \right]$$

- We can write:

$$J = J_{sT} \left[\exp \left(\frac{V_a}{kT/q} \right) - 1 \right]$$

where $J_{sT} = \left[A^* T^2 \exp \left(-\frac{\phi_{Bn}}{kT/q} \right) \right]$

- J_{sT} is the reverse saturation current density (J_{sT} is J for large reverse-bias voltage)
- Characteristics similar to PN junction diode

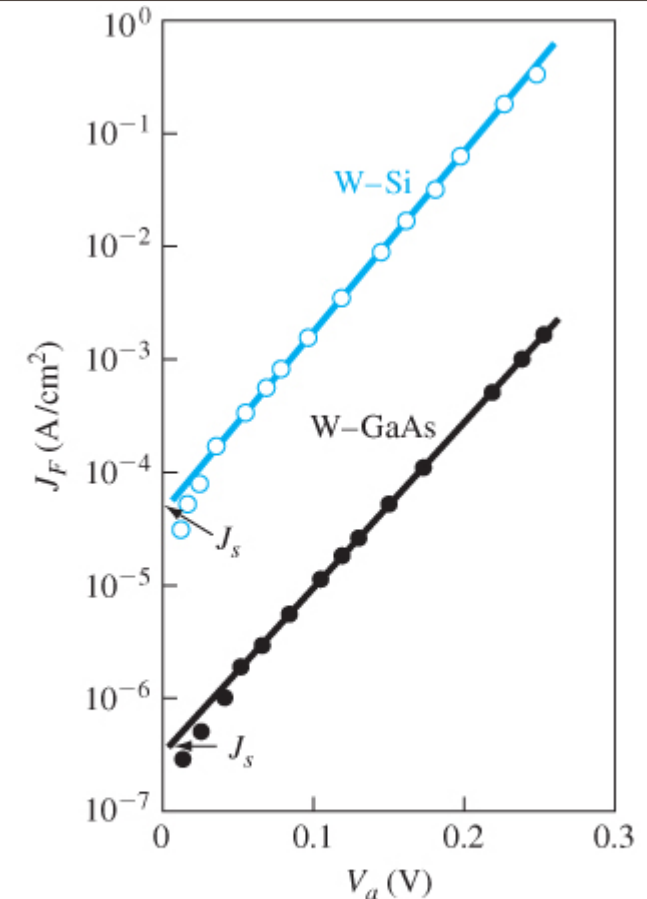


Figure 9.9 | Forward-bias current density J_F versus V_a for W-Si and W-GaAs diodes. (From Sze and Ng [15].)

Schottky Diode vs. PN Junction Diode (1)

- The form of current equation of Schottky diode and PN junction is same.
- But, there are important differences due to mechanism of current transport.

Current mechanism:

- Schottky diode: thermionic emission of majority carriers over potential barrier
- PN junction: diffusion of minority carriers

Reverse saturation current:

$$J_{sT} = \left[A^* T^2 \exp \left(-\frac{\phi_{Bn}}{kT/q} \right) \right]$$
$$J_s = \frac{qD_n n_{p0}}{L_n} + \frac{qD_p p_{n0}}{L_p}$$

- $J_{sT} \gg J_s$ [Ideal reverse saturation current orders of magnitude larger in Schottky diode]

- In real PN junction diodes generation current can dominate reverse current (generation current exists in Schottky diode also)
 - Still reverse current is two-three orders of magnitude lower in PN junction diode than J_{sT}

Schottky Diode vs. PN Junction Diode (2)

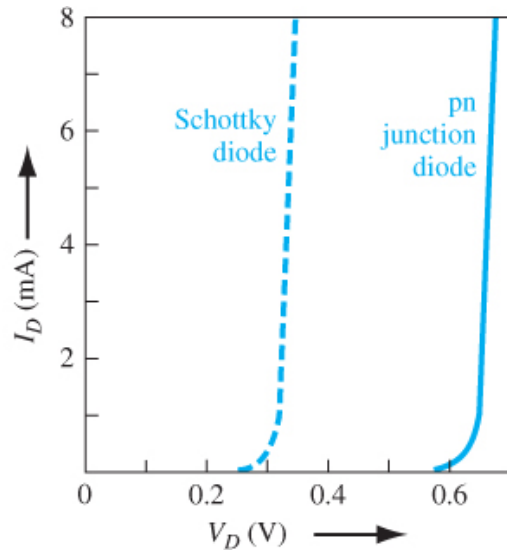


Figure 9.10 | Comparison of forward-bias I - V characteristics between a Schottky diode and a pn junction diode.

Turn-on Voltage

- Lower turn-on voltage for Schottky diodes than PN junction diode
- Exact difference depends on barrier height and doping levels

Frequency response

- No diffusion capacitance in Schottky diodes in forward-biased mode
- Schottky diodes higher frequency device than PN junction diode

Switching time from forward to reverse mode

- Schottky diodes have no minority carrier stored charge
- Schottky diodes can switch faster (switching time is in picoseconds for Schottky diodes and nanoseconds for PN junction diode)

MS Junction: Tunneling

- Width of depletion region $W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{qN_d} \right]^{1/2}$
- If N_d is very high, x_n will be very small
- Electrons can tunnel through the barrier

- Tunnelling current increases exponentially with the doping concentration

How to get good MS contact?

- Use low barrier
- Use high doping

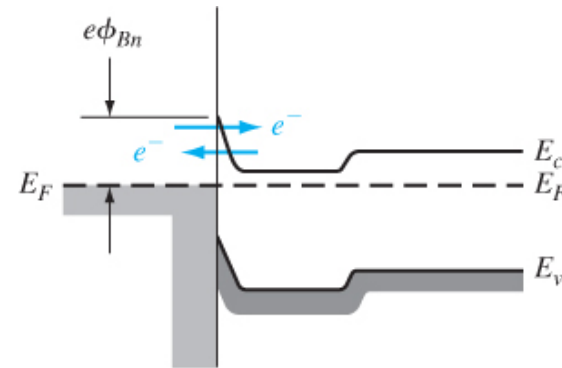


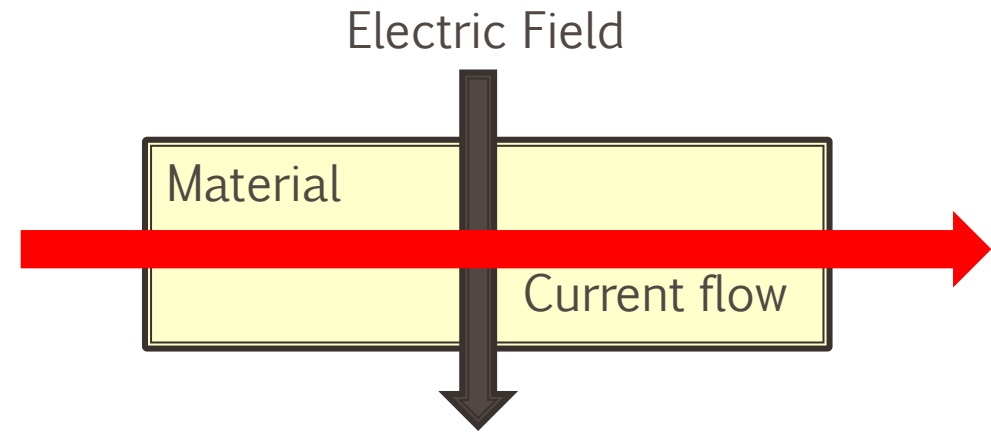
Figure 9.14 | Energy-band diagram of a heavily doped n-semiconductor-to-metal junction.

Field-Effect Transistors

Field-Effect Transistors: Basics (1)

What is field-effect?

- The phenomenon of modulating the *conductance* of a material by an *electric field applied perpendicular to its surface*.



In which type of material is field-effect observed? Metal or Semiconductor?

- Observed in semiconductors
- **Metals:** Charge density is so large that the field decays exponentially from the surface and can penetrate only a short distance into the material.
- **Semiconductors:** the penetration of the electric field far into a weakly doped semiconductor, whose charge density is not sufficient to screen the field.

Field-Effect Transistors: Structure and Operation

Structure:

- An active **channel** through which electrons (or holes) flow
- **Source** and **Drain** contacts: ohmic
- **Gate** Terminal that controls operation

Operation:

- The conductivity of the channel is modulated by voltage applied to the gate.
 - This results in the modulation of the charge density flowing in the channel.
-
- Operating principle very simple: majority carriers
 - The basic principle of MOSFET was first patented by Lilienfeld in 1925.

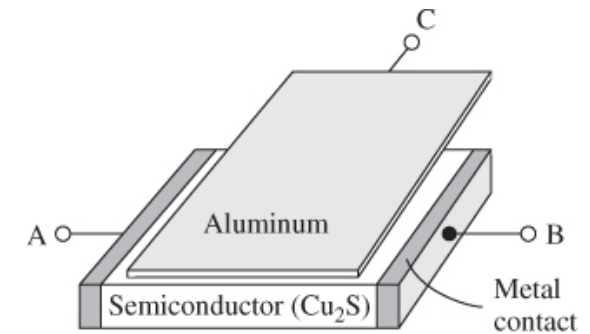
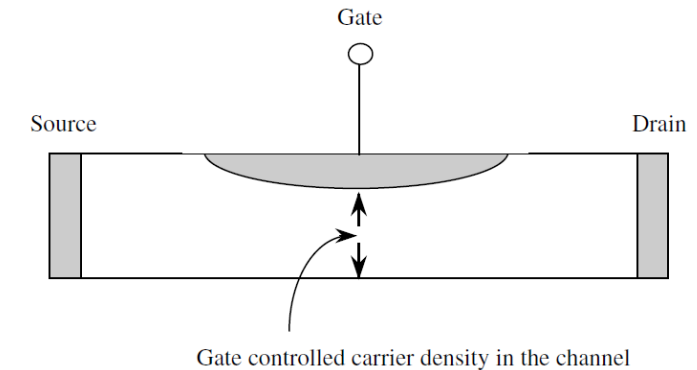
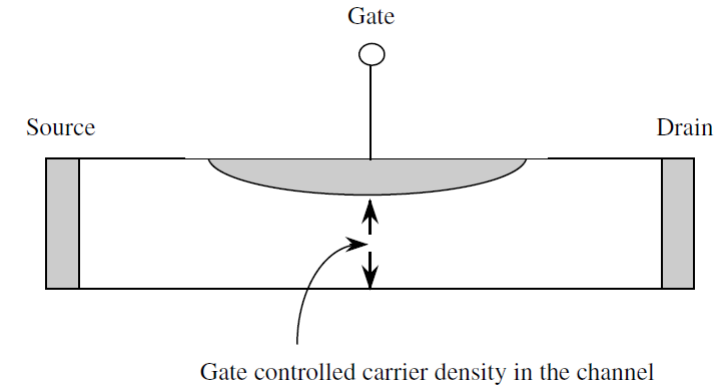


Figure 13.1 | Idealization of the Lilienfeld transistor.
(From Pierret [10].)

Field-Effect Transistors: Structure and Operation

Gate Isolation:

- It is important to isolate the gate from the channel so that no current flows into the gate (*capacitively coupled*)
- The gate isolation is done in a variety of ways, leading to a number of different devices.



Types of FET:

1. Isolation using reverse-biased PN junction: JFET
2. Isolation using MS Schottky junction: MESFET
3. Isolation using insulator/oxide: MISFET/MOSFET

- Let us first discuss all these types of FET in brief
- Then, we will go into MOSFET in depth

Field-Effect Transistors



JFET

JFET: Structure

Structure:

- **Channel:** n-type (n-channel)
- **Source:** electrons enters
- **Drain:** electrons leave
- **Gate:** p+ type creating depletion layer (two gates tied together)

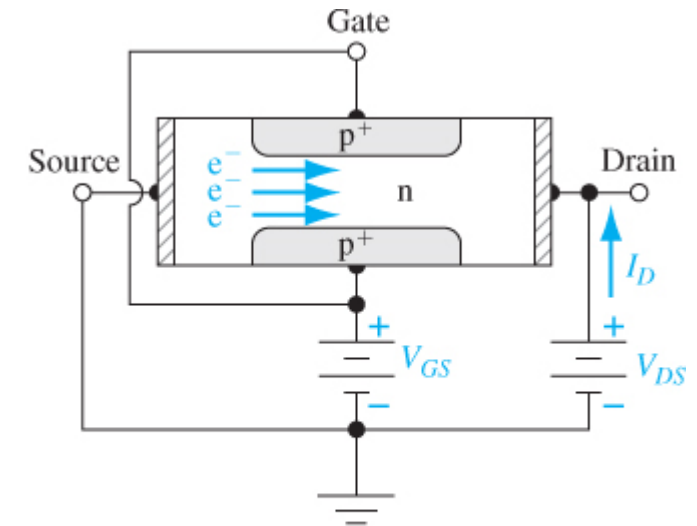


Figure 13.2 | Cross section of a symmetrical n-channel pn junction FET.

JFET: Operation (Gate Control)

Zero gate voltage:

- Channel is like a resistor
- Linear drain current with drain voltage

Small negative gate voltage:

- PN junction reverse-biased
- Depletion region widens, channel becomes narrow, resistance of channel increases

Large negative gate voltage:

- Depletion region fills the entire channel (pinch off)
- Drain current becomes zero

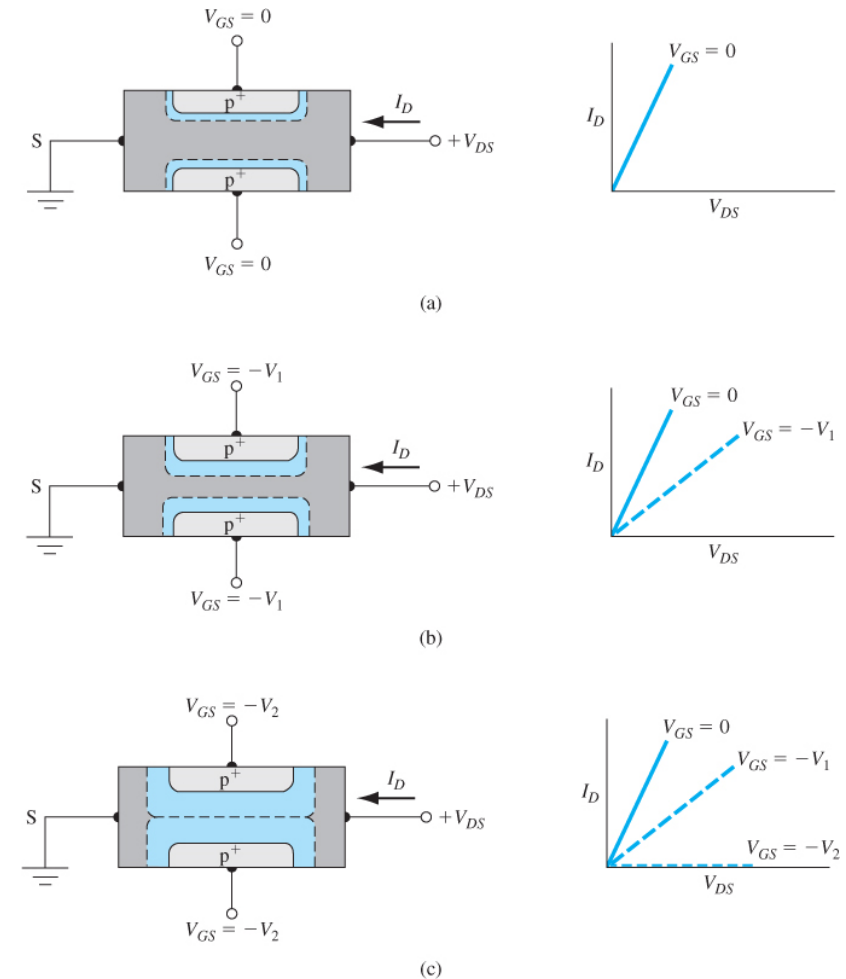


Figure 13.3 | Gate-to-channel space charge regions and I - V characteristics for small V_{DS} values and for (a) zero gate voltage, (b) small reverse-biased gate voltage, and (c) a gate voltage to achieve pinchoff.

JFET: Operation (Drain Control)

Small drain voltage (gate voltage zero):

- Channel is like a resistor
- Linear drain current with drain voltage

Increased drain voltage (gate voltage zero):

- PN junction more reverse-biased on the drain side
- Effect channel resistance increases, slope decreases

Large drain voltage (gate voltage zero):

- Channel gets **pinched off** near the drain (but still conducting on source side)
- Increased voltage do not change current (simply moves pinch off point toward source), drain current **saturates**
- Extra voltage appears across the pinched off region (that has higher resistance)

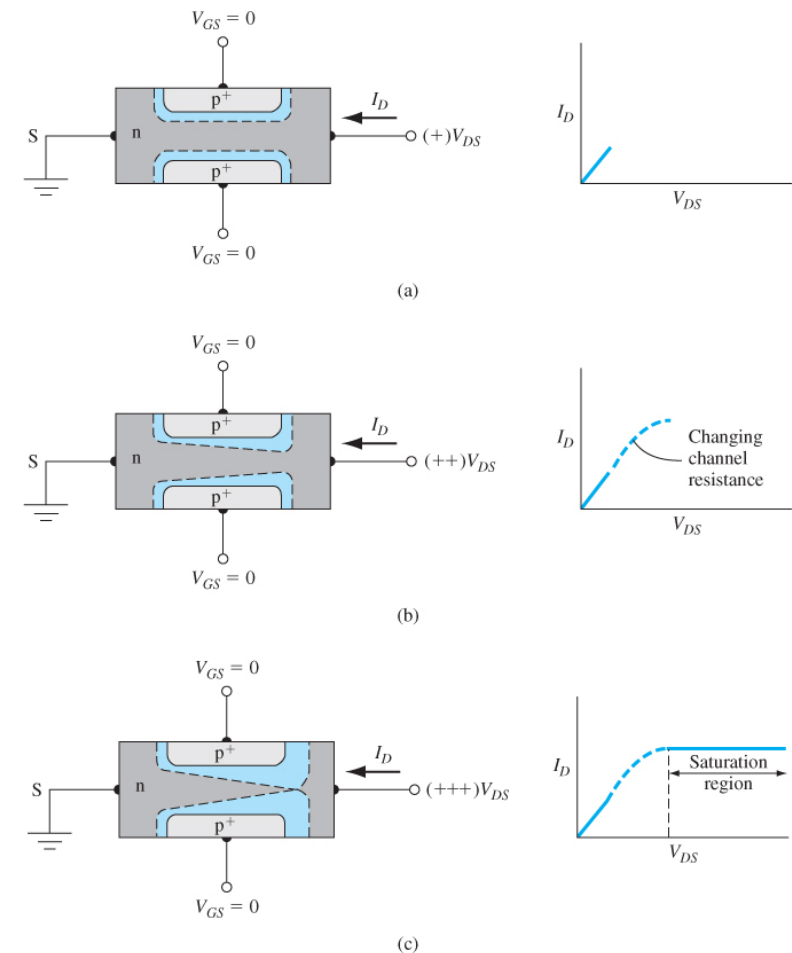


Figure 13.4 | Gate-to-channel space charge regions and I - V characteristics for zero gate voltage and for (a) a small drain voltage, (b) a larger drain voltage, and (c) a drain voltage to achieve pinchoff at the drain terminal.

Field-Effect Transistors



MESFET

MESFET: Structure and Operation

Structure:

- **Channel:** n-type (n-channel)
 - **Source:** electrons enters (Ohmic)
 - **Drain:** electrons leave (Ohmic)
 - **Gate:** Rectifying Schottky contact
-
- GaAs typically used for MESFET because of higher mobility
 - Operation similar to JFET
 - Normally ON, and channel gets depleted by reverse-biased Schottky junction
 - Key: how to design Schottky junction to limit gate current

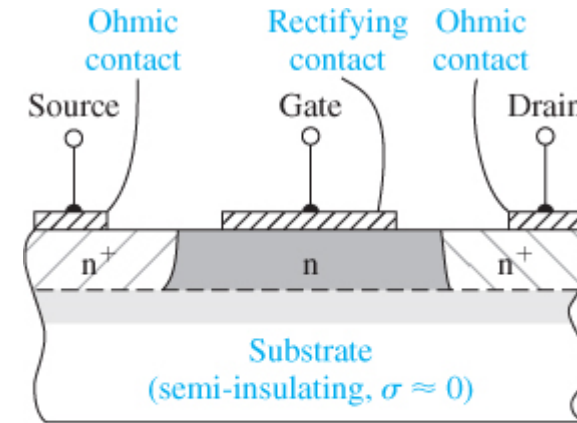
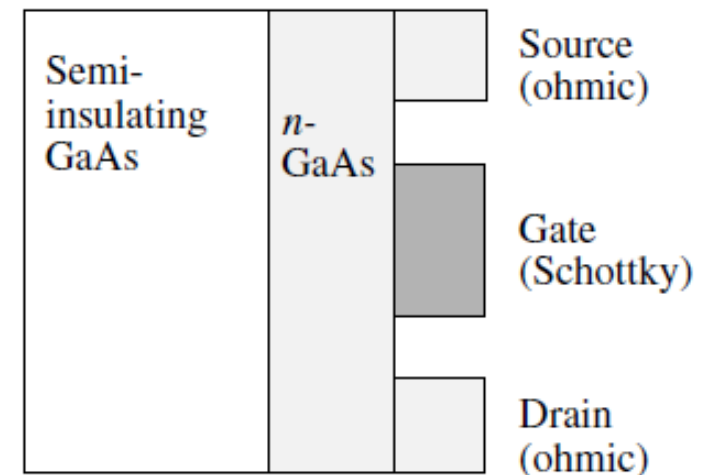


Figure 13.6 | Cross section of an n-channel MESFET with a semi-



References

- D. A. Neaman, *"Semiconductor Physics and Devices"*